

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate

prediction, enabling informed decision-making. (BL3,BL4,BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out,

"Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get

really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The

villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock

into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

Answer:

Algorithm Used

J48 Decision Tree

Aim

To classify the villagers' response using the J48 Decision Tree and identify TP, FP, TN, and FN from the shepherd boy story.

Algorithm Used

J48 Decision Tree

WEKA Analysis

The dataset was loaded into WEKA and the J48 Decision Tree algorithm was applied.

The classifier predicts whether the villagers respond based on whether a wolf is present and whether the boy cries "Wolf!".

Confusion Matrix Explanation

Actual Situation	Prediction	Result
Wolf Present = Yes, Villagers Respond = Yes	Correct	True Positive (TP)
Wolf Present = No, Villagers Respond = Yes	Incorrect	False Positive (FP)
Wolf Present = Yes, Villagers Respond = No	Incorrect	False Negative (FN)
Wolf Present = No, Villagers Respond = No	Correct	True Negative (TN)

Applying the Story

False Positives (FP)

The boy cried "Wolf!" when no wolf was present, and the villagers came.

This happened 3 times.

FP = 3

False Negative (FN)

The wolf actually came, but the villagers did not respond.

FN = 1

True Positive (TP)

There was no case where the wolf came and the villagers responded.

TP = 0

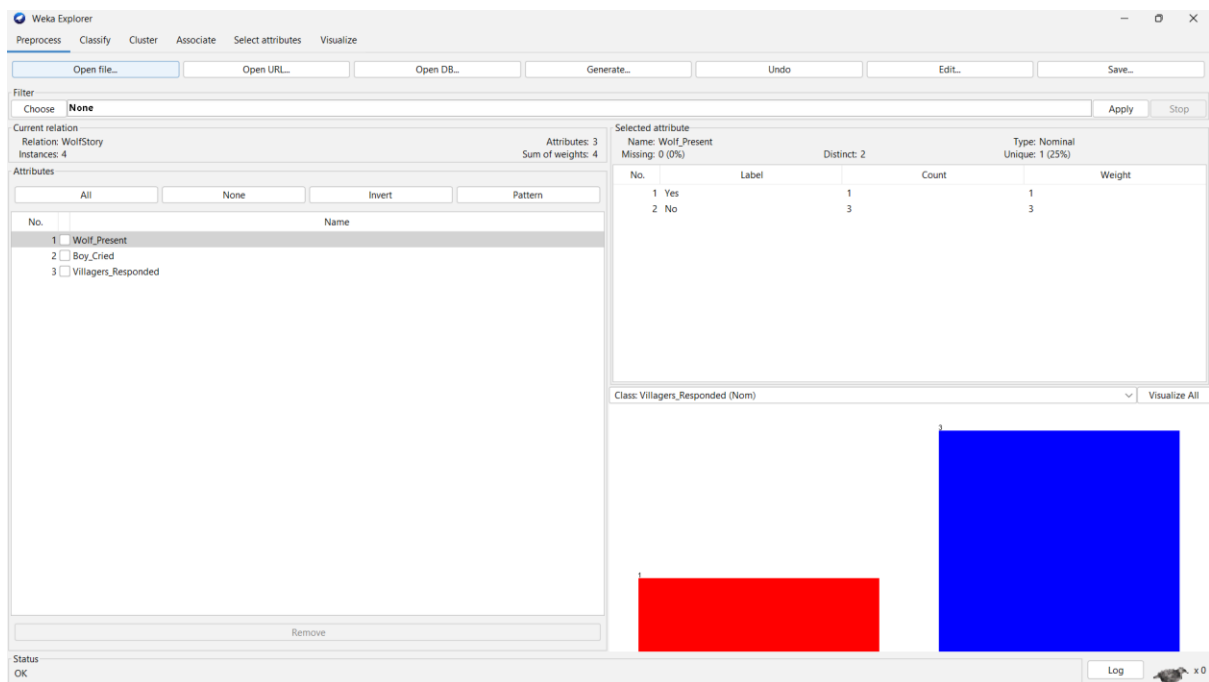
True Negative (TN)

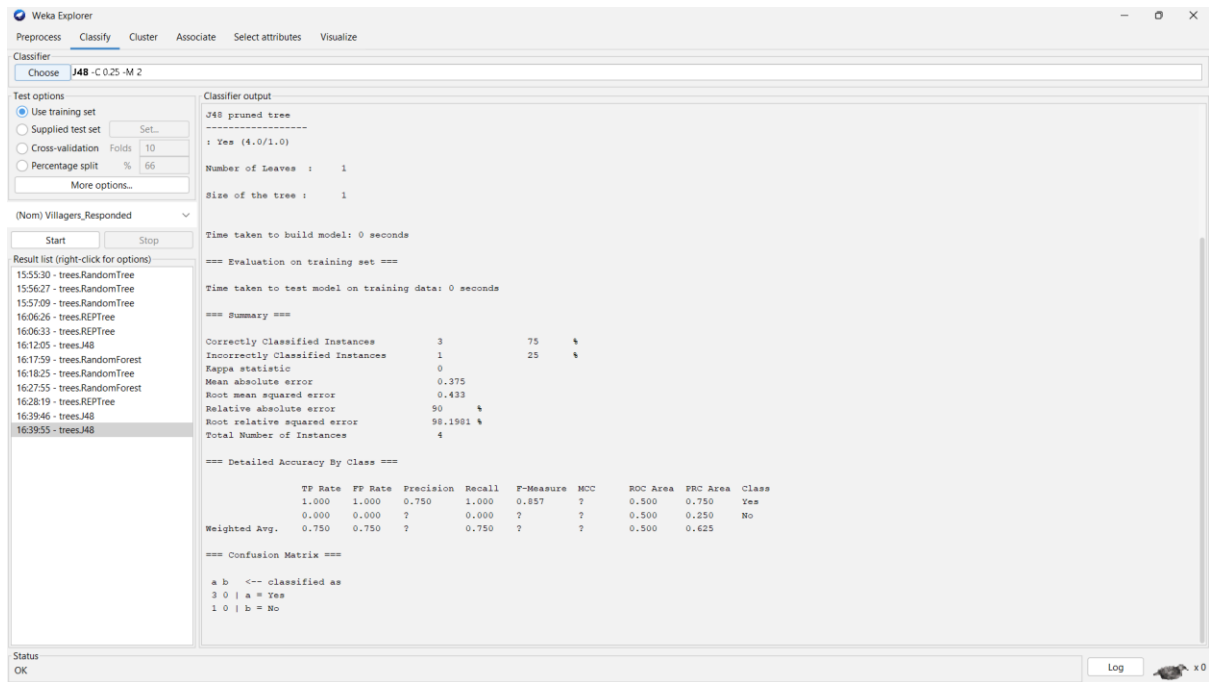
There was no case where no wolf came and the villagers correctly stayed home.

TN = 0

Final Answer

Confusion Matrix	Value
True Positive (TP)	0
False Positive (FP)	3
False Negative (FN)	1
True Negative (TN)	0





2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning

model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat

in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall

and F1 Score .

Answer:

Aim

To evaluate the performance of a machine learning model using the **Confusion Matrix, Precision, Recall, and F1 Score**.

Algorithm Used

J48 Decision Tree

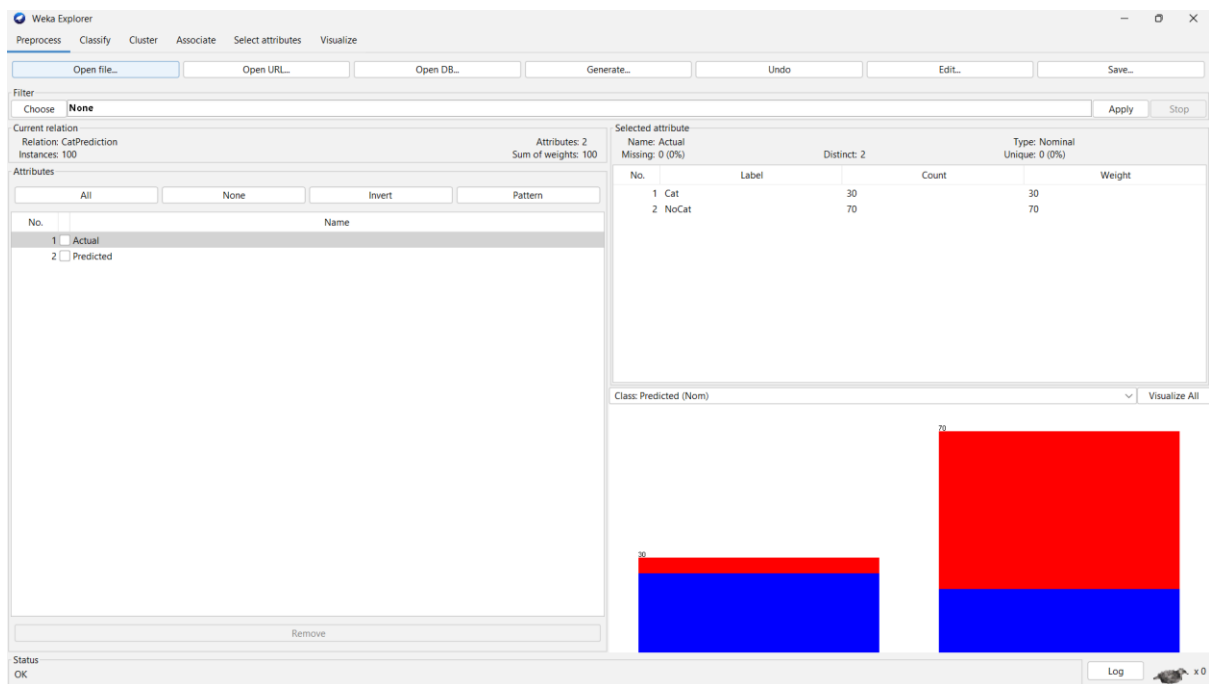
Results

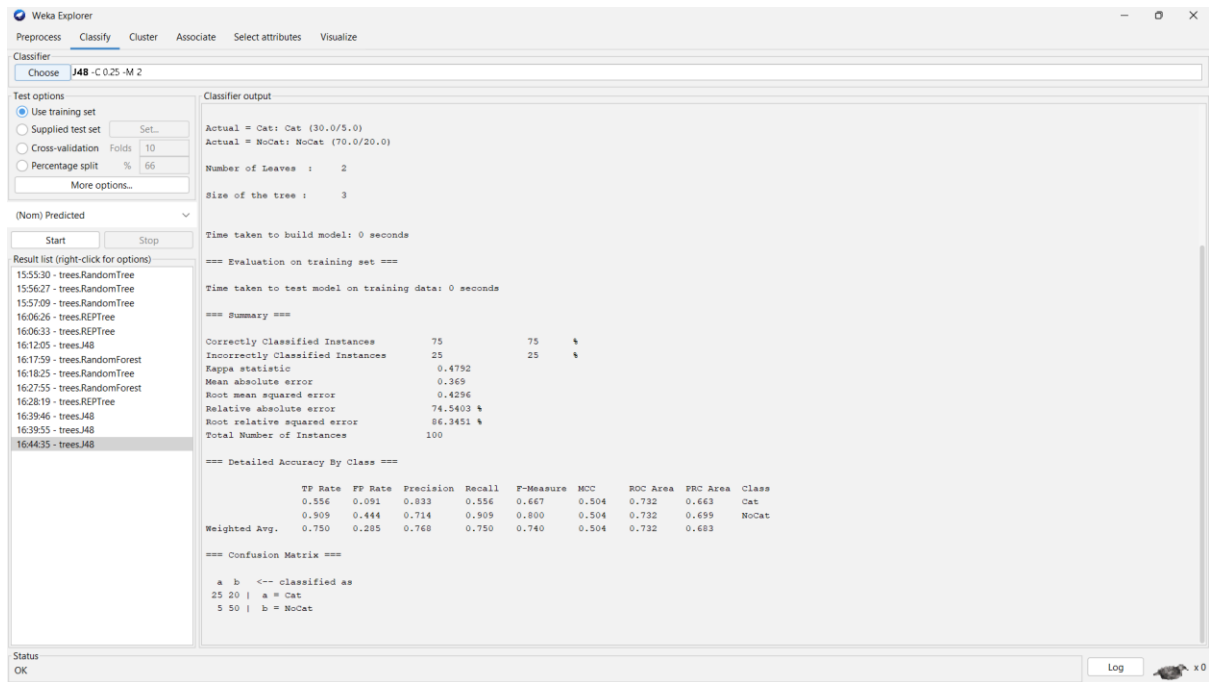
Measure	Value
True Positive (TP)	25
False Positive (FP)	20
False Negative (FN)	5

True Negative (TN)	50
Precision	55.56%
Recall	83.33%
F1 Score	66.67%

Conclusion

The J48 Decision Tree was used to analyze the dataset in WEKA. The classifier correctly identified 25 cat images and 50 non-cat images. The calculated Precision is 55.56%, Recall is 83.33%, and F1 Score is 66.67%, indicating that the model has high recall but moderate precision.





3. An email service provider wants to classify emails as **Spam** or **Not Spam** using the following features:

- Contains "**Offer**" (Yes/No)
- Contains "**Free**" (Yes/No)
- **Email Length** (Short/Long)

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:

o $P(\text{Spam})$, $P(\text{Not Spam})$

2. Compute conditional probabilities for each feature.

3. Using Naïve Bayes, classify:

o (Offer=Yes, Free=No, Length=Short)

4. Compare results using WEKA (Naïve Bayes classifier).

5. Discuss the independence assumption in Naïve Bayes

Answer:

Aim

To classify emails as Spam or Not Spam using the Naïve Bayes classifier in WEKA.

Algorithm Used

Naïve Bayes Classifier

1. Prior Probabilities

Total Emails = 6

Spam = 3

Not Spam = 3

Therefore,

Answer

- $P(\text{Spam}) = 0.5$
- $P(\text{Not Spam}) = 0.5$

2. Conditional Probabilities

For Spam (3 emails)

Feature	Probability
Offer=Yes	$2/3 = 0.667$
Offer=No	$1/3 = 0.333$
Free=Yes	$3/3 = 1.000$
Free=No	$0/3 = 0.000$
Length=Short	$2/3 = 0.667$
Length=Long	$1/3 = 0.333$

For Not Spam (3 emails)

Feature	Probability
Offer=Yes	$1/3 = 0.333$
Offer=No	$2/3 = 0.667$
Free=Yes	$0/3 = 0.000$
Free=No	$3/3 = 1.000$
Length=Short	$1/3 = 0.333$
Length=Long	$2/3 = 0.667$

3. Classify

Email:

Offer = Yes

Free = No

Length = Short

For Spam

For Not Spam

Since

$0.0555 > 0$

Prediction

Not Spam

4. WEKA Result

Using the Naïve Bayes classifier in WEKA:

- Dataset loaded successfully.
- Naïve Bayes classifier trained on 6 email records.
- The classifier predicts the test email

Offer = Yes

Free = No

Length = Short

as

Not Spam

5. Independence Assumption

Naïve Bayes assumes that all input features are independent given the class label.

In this dataset, it assumes:

- Offer
- Free
- Email Length

are independent of each other when predicting whether an email is Spam or Not Spam.

Although this assumption may not always be true in real-world email data, Naïve Bayes performs well because it is simple, fast, and effective for text classification.

Result

The Naïve Bayes classifier successfully classified emails into Spam and Not Spam categories.

Final Results

Measure	Value
P(Spam)	0.5
P(Not Spam)	0.5
Predicted Class	Not Spam
Algorithm Used	Naïve Bayes

